

Aiden Zhou

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EDUCATION

Yale University

Expected: **May 2028**

Bachelor of Science in **Computer Science** and **Mathematics**

GPA: **3.93/4.0**

- **Relevant Coursework:** Algorithms, Data Structures, Probability Theory, Advanced Linear Algebra, Real Analysis, Discrete Math, Computational Vision, Graph Deep Learning, Fourier Analysis
- **Leadership:** Multimedia Managing Editor of the *Yale Daily News*, Features Editor of the *Yale Scientific Magazine*, Teaching Assistant for Calculus I/II and Real Analysis

EXPERIENCE

Cohere | Machine Learning Intern

May 2026 – Present

- Machine Learning Intern on Cohere's Sovereign AI team, contributing to data infrastructure and foundational model post-training (SFT, RLVR) for enterprise and government applications

Supervised Program for Alignment Research (SPAR) | Research Intern

Feb 2026 – May 2026

- Developed adaptive benchmarking methods for LLM safety evaluation, reducing cost by 99.7% and preserving true model rankings across six benchmarks; paper accepted at AI4GOOD @ ICML 2026
- Integrated and deployed three Pyro / py-irt response model variants for ordinal and continuous outcomes, expanding adaptive evaluation to agentic tasks and richer scoring regimes

Wu Tsai Institute at Yale University | Research Intern

Jan 2025 – Oct 2025

- Engineered and tested inverse graphics vision models that infer 3D face shape and texture in one feedforward pass, matching human neural representations across three hierarchical processing stages
- Built an end-to-end ETL pipeline to clean and regularize 50+ GB of ECoG data from 60+ patients across 85 tasks; ran CUDA/Slurm GPU workflows and cut manual workload by ~5x

AlgoVerse AI | Machine Learning Researcher

May 2025 – Aug 2025

- Discovered persistent trends in learning dynamics and neuron polysemanticity throughout language model pretraining; presented first-author paper at MechInterp @ NeurIPS 2025

PROJECTS

Adaptive Interaction Graphs for Particle Simulation (github.com/aidenzhou8/AdaptGNS)

- Developed AdaptGNS, a graph-based particle simulator that adaptively expands uncertain regions to improve long-horizon rollout accuracy; sole-author paper accepted at AI4Physics @ ICML 2026

Graded LLM Evaluation with Item Response Theory (github.com/aidenzhou8/Ordinal-FBM)

- Extended Fluid Benchmarking beyond binary outcomes to support ordinal and continuous IRT methods such as GRM and GPCM, enabling efficient evaluation for a broader class of benchmarks

NeuronEvolution: Tracking LLM Features over Pre-Training (neuronevolution.streamlit.app)

- Deployed a web app to track feature evolution across 143 checkpoints of Pythia LLMs; implemented Python-based backend and used Snowflake and AWS to ingest model activations into a data warehouse

SKILLS

- **Languages:** Python, C/C++, Java, JavaScript, MATLAB, SQL, HTML/CSS, Bash, Racket
- **Machine Learning:** PyTorch, scikit-learn, PyG, Transformers, CUDA, vLLM, Pyro, py-irt, RLVR
- **Systems / Infrastructure:** AWS, GCP, Git, Docker, Kubernetes, Weights & Biases, Grafana, Slurm

HONORS AND AWARDS

- **Yale First-Year Summer Research Scholarship in the Sciences and Engineering (2025)**
- **Math and Science Olympiads:** AIME (3x), CMO, UBC Physics Gold (2x)
- **Chess:** US National Master; 4th at the 2022 North American Youth Championship, 34th at the 2022 World Youth Championship, and top-30 finishes at the World Open and North American Open